

Thermoelectric Performance Boost of n-Type Carbonized Polydopamine through Nanoarchitectonics with Synergistic Catalysis and Coordination Strategy

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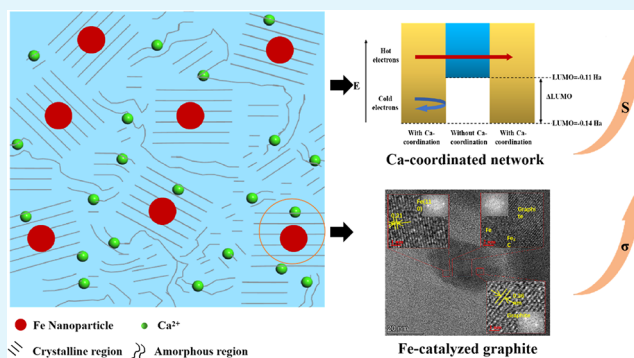
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Supporting Information

ABSTRACT: The lagging development of n-type organic thermoelectric materials based on a conjugated polymer drastically curtails the advancement of high-performance flexible thermoelectric devices, which is largely attributed to the lack of effective doping approaches. Bioinspired polydopamine (PDA) materials possess abundant functional groups that provide a platform for doping and other functional processes. In this study, we developed a strategy that synergistically integrates calcium (Ca) coordination and iron (Fe) nanoparticle catalysis during the carbonization process of PDA to enhance the thermoelectric performance of n-type carbonized-PDA (cPDA). Fe nanoparticles catalyzed the growth of graphite nanocrystals in cPDA during the carbonization process, resulting in a conductivity enhancement from 6.9×10^4 to 3.31×10^5 S m⁻¹. Meanwhile, Ca coordination introduced a lower energy level of the lowest-unoccupied molecular orbital (LUMO) orbit, selectively supplying high-energy electrons to boost the Seebeck coefficient from -4.22 to -10.47 μ V K⁻¹. As a result, the power factor (PF) increased nearly 30 times, and a PF value of 36.3 μ W m⁻¹ K⁻² was recorded. Furthermore, the improved cPDA materials are thermally stable up to 500 °C. This work reveals the potential of cPDA in n-type thermoelectric applications and provides a route for the development of high-performance flexible thermoelectric materials.

KEYWORDS: N-type materials, thermoelectric properties, carbonized polydopamine, Ca coordination, nanoarchitectonics



1. INTRODUCTION

Organic flexible thermoelectric materials demonstrate unique application prospects in portable electronics, smart textiles, and biomedical sensors due to their low-cost processing, environmental friendliness, wearable compatibility, and ultralight characteristics.^{1–6} The thermoelectric conversion efficiency is quantified by the dimensionless figure of merit $ZT = S^2\sigma T/\kappa$ (where S represents the Seebeck coefficient, σ represents the electrical conductivity, κ represents the thermal conductivity, and T represents the absolute temperature), with the power factor ($PF = S^2\sigma$) serving as the primary evaluation metric for organic thermoelectric performance. Given organic materials' inherently low κ , performance optimization predominantly focuses on PF enhancement. Effective thermoelectric devices rely on complementary pairs of p- and n-type materials. To date, organic p-type thermoelectric materials have achieved remarkable advancements, for instance, a PF value of above 500 μ W m⁻¹ K⁻² in the p-type polymer poly(3,4-ethylenedioxythiophene) (PEDOT) was achieved.^{7–9} However, the development of n-type organic thermoelectric materials has been significantly hindered. Most of the n-type conjugated

polymer thermoelectric materials exhibit σ values less than 2000 S m⁻¹ and PF values less than 10 μ W m⁻¹ K⁻².¹⁰ The poor performance of n-type organic thermoelectric materials is mainly constrained by the environmental stability and the lack of effective n-type dopants.¹¹

Bioinspired polydopamine (PDA) and its derivatives have recently attracted the attention of researchers because of their unique structure and regulatory properties.¹² PDA's abundant catechol/amino functional groups provide exceptional adhesion and surface engineering adaptability.^{13–18} After high-temperature carbonization, the resulting carbonized-PDA (cPDA) exhibits short-range ordered nitrogen-doped graphitic nanocrystalline structures.^{19,20} Nanocrystals composed of two-dimensional honeycomb-like sp²-hybridized lattices maintain

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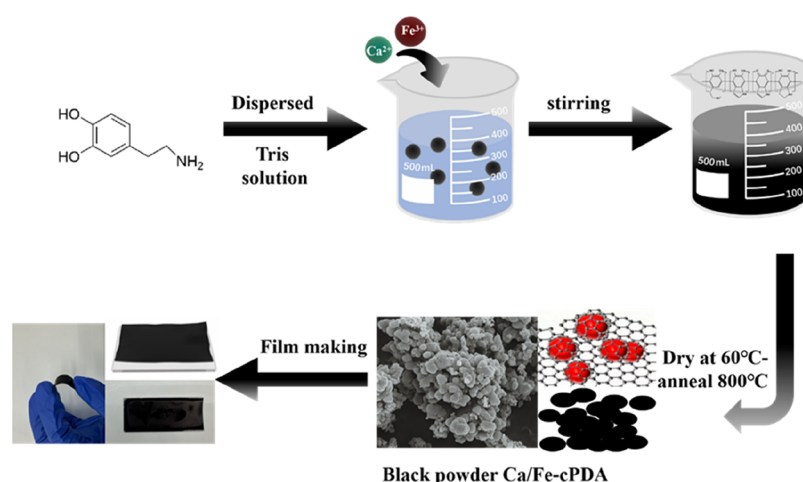


Figure 1. Diagram for the synthesis of Ca/Fe-cPDA.

stable carrier transport even under extreme conditions including high temperatures and mechanical stress. The long-range disordered amorphous matrix surrounding them allows interconnection via quantum tunneling and hopping conduction mechanisms. Compared to conventional n-type polymers that rely on complete molecular chain conjugation (vulnerable to interchain steric hindrance and inhomogeneous doping), cPDA demonstrates reliable transport stability in complex environments.^{21–23} However, the reported cPDA materials so far remain constrained by a low graphitization degree, with PF still limited to low values ($\sim 2 \mu\text{W m}^{-1} \text{K}^{-2}$).

Wang et al. achieved the highest Seebeck value of $-30.2 \mu\text{V K}^{-1}$ and a maximum power factor value of $35.57 \mu\text{W m}^{-1} \text{K}^{-2}$ through the fabrication of the c-CNT/PDA composite. However, the charge transport in this composite is predominantly governed by the CNT component, with PDA primarily serving as an interfacial junction layer between individual components.²⁴ Li et al. doped cPDA with Na⁺, Mg²⁺, and Cu²⁺ ions, yet the resulting cPDA materials exhibited no significant improvement in either electrical conductivity or Seebeck coefficient.²⁵ Fortunately, recent studies have demonstrated that metal-ion coordination can enhance n-type thermoelectric performance of polymers through orbital hybridization between metal d-orbitals and polymer molecular orbitals.^{26–28} On the other hand, Fe nanoparticles can serve as efficient catalysts for graphitic carbon growth through accelerating sp²-hybridized carbon domain formation, providing a viable pathway to enhance the graphitization degree of cPDA.^{29–31} Therefore, by leveraging the abundant functional groups of PDA, we engineered a synergistic strategy integrating Ca ion coordination and Fe nanoparticle catalysis to enhance the thermoelectric performance of cPDA in this work. The efficacy of this strategy is corroborated by the substantially enhanced thermoelectric performance (S , σ , and PF) and a comprehensive suite of structural characterizations, including X-ray diffraction (XRD), X-ray photoelectron spectroscopy (XPS), Raman spectroscopy, and transmission electron microscopy (TEM) analyses. Fe nanoparticles catalyze graphitic nanocrystal formation during pyrolysis, resulting in a high σ of $3.31 \times 10^5 \text{ S m}^{-1}$. Density functional theory (DFT) modeling indicates that Ca ion coordination creates the energy-filtering effect to boost S to $-10.47 \mu\text{V K}^{-1}$ in cPDA. As a result, the power factor of conventional PDA-based thermoelectric composites ranges

from about 1.7 to $14.4 \mu\text{W m}^{-1} \text{K}^{-2}$ (Table S1), while Ca/Fe-cPDA obtained a power factor of $36.3 \mu\text{W m}^{-1} \text{K}^{-2}$, which is a 2.52-fold increase in PF compared to the highest PDA-based complex. This is mainly due to the synergistic effect of increased Fe-induced graphitization and the energy-filtering effect induced by calcium ion coordination. The cPDA materials also demonstrate thermal stability at a temperature of 500 °C, providing new possibilities for extreme-environment applications of flexible thermoelectric devices.

2. EXPERIMENTAL SECTION

2.1. Sample Preparation. Dopamine (DA) hydrochloride powder (99%, Alfa Aesar) and tris(hydroxymethyl) aminomethane (Tris, + 99.9%, Alfa Aesar) were used as received. Hydrochloric acid (HCl, AR), poly(vinyl alcohol) (PVA, ~ 1799 , average polymerization degree is 1700 ± 50 , degree of alcoholysis is 98.0–99.0%, viscosity is 25.0–31.0 mPa·s), CaCl₂, and FeCl₃·6H₂O were also commercially purchased. PDA was synthesized by dissolving DA hydrochloride (1 g) in 500 mL of tris-HCl buffer solution (pH 8.5) under continuous mechanical stirring at room temperature for 48 h to allow spontaneous polymerization. The resulting PDA precipitate was collected, washed with deionized water, and dried at 60 °C. For metal-incorporated composites, stoichiometric amounts of inorganic salts (CaCl₂ and FeCl₃·6H₂O) were simultaneously added to the tris-HCl buffer solution with DA hydrochloride. Ca-coordinated cPDA and Fe-catalyzed cPDA were prepared by introducing Ca²⁺ or Fe³⁺ ions alone, while Ca/Fe-cPDA composites with varying Ca²⁺:Fe³⁺ molar ratios (1:1, 1:2, and 2:1) were synthesized by adjusting the salt proportions. The molar ratio of (Ca+Fe):DA hydrochloride was kept at 1:10 for all Ca/Fe-cPDA composites. All mixtures underwent identical polymerization and purification procedures as described for pure PDA. The as-prepared PDA and metal-incorporated PDA powders were carbonized at 800 °C for 4 h under a nitrogen atmosphere (heating rate: 5 °C/min) to obtain cPDA, Ca-coordinated cPDA, Fe-catalyzed cPDA, and Ca/Fe-cPDA powder composites. 1 mL of ethanol and 4 mL of deionized water were mixed well, and 0.1 g of carbon material was added. The composite material was treated with ultrasound for 1 h and vibrated with a cell crusher for 30 min, and a uniform dispersion of the carbon material in the solution could be observed. Then, 3 mL of poly(vinyl alcohol) was added and stirred at low speed for 20 min. Homogeneous drops were placed on slides, placed in an oven (60 °C) for 24 h, and removed for thermoelectric properties testing.

2.2. Characterization Methods. An X-ray diffractometer (XRD, Philips X'PertMRD, Netherlands) was employed to study the crystal structure of the samples with a scanning rate of 5° min^{-1} using Cu K α radiation. Raman spectra were recorded on a Raman spectrometer with an excitation wavelength of 532 nm. X-ray photoelectron

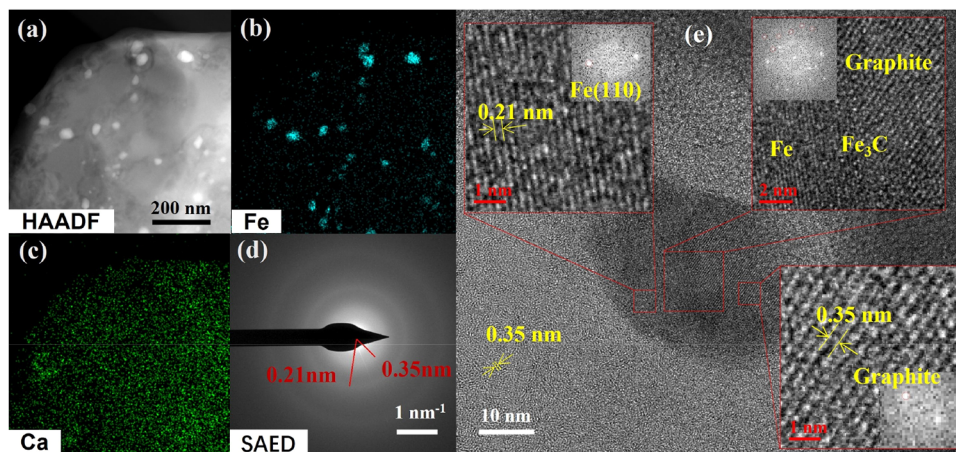


Figure 2. Morphological analysis of cPDA synthesized with Fe^{3+} and Ca^{2+} . (a) HAADF image. (b, c) Corresponding EDS mapping of Fe and Ca elements. (d) SAED pattern. (e) HRTEM image.

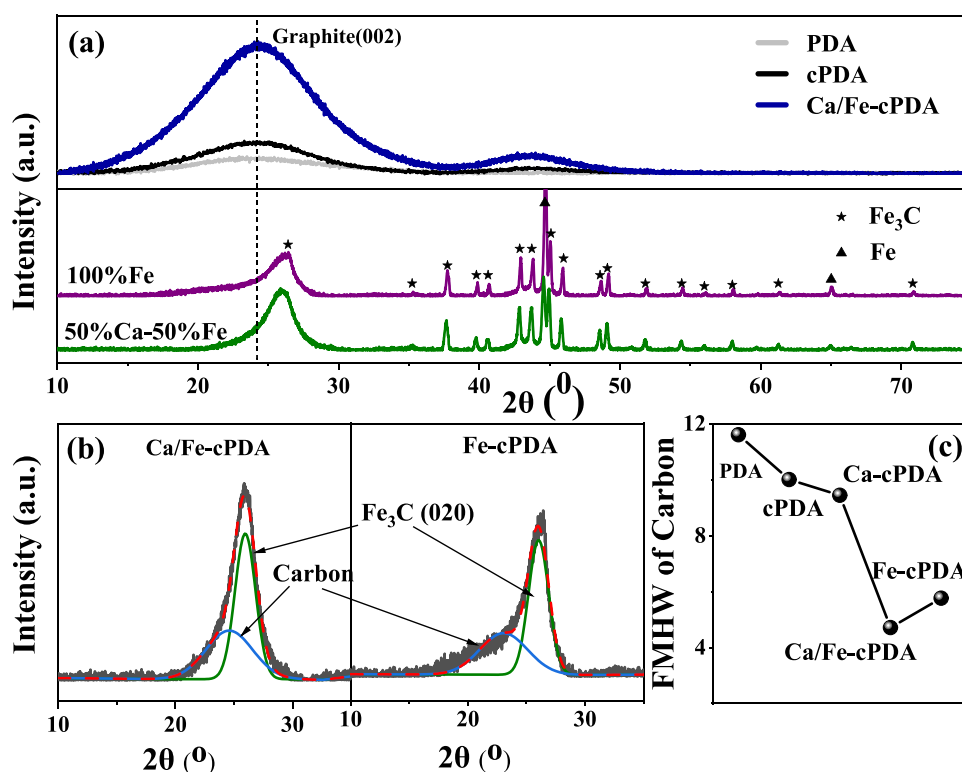


Figure 3. Structural analysis of PDA and cPDA synthesized with Fe^{3+} and Ca^{2+} . (a) XRD patterns. (b) Fitting results of carbon and Fe_3C in Fe-catalyzed cPDA. (c) Fwhm of carbon in cPDA synthesized by different processes.

spectroscopy (XPS, ULVAC-PHI, Japan) was used to analyze the chemical composition of the samples' surfaces. Transmission electron microscopy (JEOL F200, United States) was employed to examine the microstructure of the samples. A field emission scanning electron microscope (FE-SEM, Zeiss Sigma) equipped with energy-dispersive X-ray spectroscopy (EDS, Xplore, Oxford Instruments) was subsequently used to detect the chemical compositions of these samples. The details of materials characterization, thermoelectric measurement, and computational methods are presented in the [Supporting Information](#). The Material Studio package was used for the electronic structure simulations.

3. RESULTS AND DISCUSSION

3.1. Synthesis of $\text{Ca}^{2+}/\text{Fe}^{3+}$ Codoped cPDA. Figure 1 presents a schematic of the n-type Ca/Fe-cPDA composite film

preparation. The DA monomer and metal salt in an alkaline solution self-polymerize via mechanical stirring at room temperature. After centrifugal cleaning, drying, and high-temperature ($800\text{ }^\circ\text{C}$) annealing, black cPDA powder formed. Dropping the cPDA mixed solution onto a slide yielded the composite film.

3.2. $\text{Ca}^{2+}/\text{Fe}^{3+}$ Codoping Effect on Graphitic Nano-crystal in cPDA. Figure 2 presents the TEM images of cPDA synthesized with Fe^{3+} and Ca^{2+} . The high-angle annular dark-field (HAADF) image in Figure 2a reveals the precipitation of metal nanoparticles in cPDA, which is represented by the bright spots. The corresponding energy-dispersive X-ray spectroscopy (EDS) mappings in Figure 2b,c confirm the precipitation of Fe nanoparticles, while the Ca element is

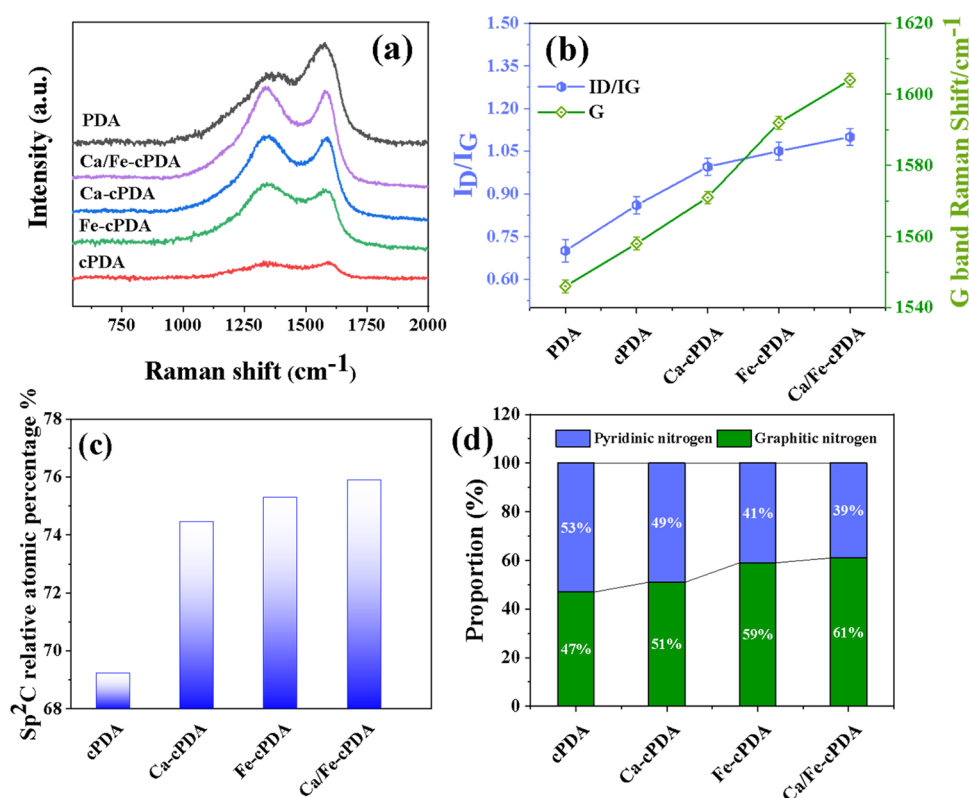


Figure 4. Raman spectra and XPS results of cPDA and cPDA synthesized with Fe³⁺ and Ca²⁺. (a) Raman spectra. (b) I_D/I_G and G-band shift values. (c) sp² C relative atomic percentages. (d) Proportion of pyridinic nitrogen and graphitic nitrogen in N 1s spectra.

scattered everywhere. X-ray photoelectron spectroscopy (XPS) measurement in Figure S1 shows the characteristic Ca 2p_{3/2} peaks at 347.01 eV (Ca-cPDA) and 347.15 eV (Ca/Fe-cPDA), which indicates that the Ca element is in the +2 valence state.³² The 0.14 eV shift in Ca/Fe-cPDA is attributed to partial electron transfer during Fe³⁺ reduction. However, no valid Fe 2p peaks are detected. These results indicate that Ca²⁺ ions are successfully coordinated within the cPDA, while Fe³⁺ ions undergo reduction to form Fe nanoparticles during the high-temperature carbonization process. Furthermore, the selected area electron diffraction (SAED) patterns in Figure 2d reveal two distinct rings indexed to *d*-spacings of 0.35 and 0.21 nm, corresponding to the graphite (002) plane and the Fe(110) plane, respectively. High-resolution transition electron microscopy (HRTEM) in Figure 2e provides direct evidence of the carbonized structure of cPDA synthesized with Fe³⁺. The microstructure predominantly manifests as a short-range ordered amorphous carbon composed of numerous graphene-like layered domains with 2–3 layers stacking and an interlayer spacing of approximately 0.35 nm. Charge carrier transport occurs via a hopping mechanism.^{33–35} A large grain predominantly composed of Fe nanoparticles and graphite was observed. The TEM image in Figure S2 reveals that the particle size of Fe nanoparticles is roughly 30 to 50 nm. The interlayer spacing of the left grain, measured as 0.21 nm via inverse fast Fourier transform (FFT) analysis, is assigned to the Fe(110) crystallographic plane. In contrast, the right grain exhibits an interlayer spacing of 0.35 nm, consistent with the theoretical plane spacing of graphene. Meanwhile, FFT confirms the existence of a polycrystalline structure at the interface between the Fe nanocrystal and graphite, which is likely primarily composed of iron carbide. Therefore, the

precipitated Fe nanocrystal catalyzed the graphitization of amorphous carbon during the high-temperature carbonization.^{29–31} The intermediate iron carbide phase acts as a structural template for carbon atom alignment, while facilitating graphitic nucleation through lattice matching. Notably, the Fe nanoparticles exhibit lateral asymmetry in graphitization, with well-defined graphite crystalline exclusively developing on the right facets, whereas the left interfaces retain amorphous carbon due to the different catalytic activity on different crystal faces.³⁶ The additional HRTEM images of multiple Fe nanocrystalline particles in Figure S3 further confirm that graphite-like nanocrystals exhibit higher occurrence rates on the Fe(110) surfaces.

X-ray diffraction (XRD) patterns can corroborate the evolution of a graphite-like structure in cPDA synthesized with Fe³⁺. Figure 3a shows the XRD patterns of PDA and cPDA synthesized with Fe³⁺ and Ca²⁺. All of the cPDA powders exhibit broad diffraction peaks around 24 and 44°, indicating an amorphous carbon frame. As for the cPDA synthesized with Fe³⁺, Fe, and Fe₃C (cementite), precipitation is exhibited. Moreover, the peaks of amorphous carbon and Fe₃C (020) overlap, creating an asymmetric peak. The fitting results of the spectra in the 2θ range of 20–30° are shown in Figure 3b. Thereafter, the crystallinity degree of the cPDA powders is evaluated through the calculated full width at half-maximum (fwhm) of the amorphous carbon peak in Figure 3c. Compared with PDA, the amorphous carbon peak of cPDA showed a narrowed fwhm due to the carbonization during the high-temperature annealing. After carbonization with Ca²⁺, its fwhm exhibited a slight decrease. Notably, the cPDA synthesized with Fe³⁺ shows a sharp reduction in the fwhm of the amorphous carbon peak, demonstrating a significant

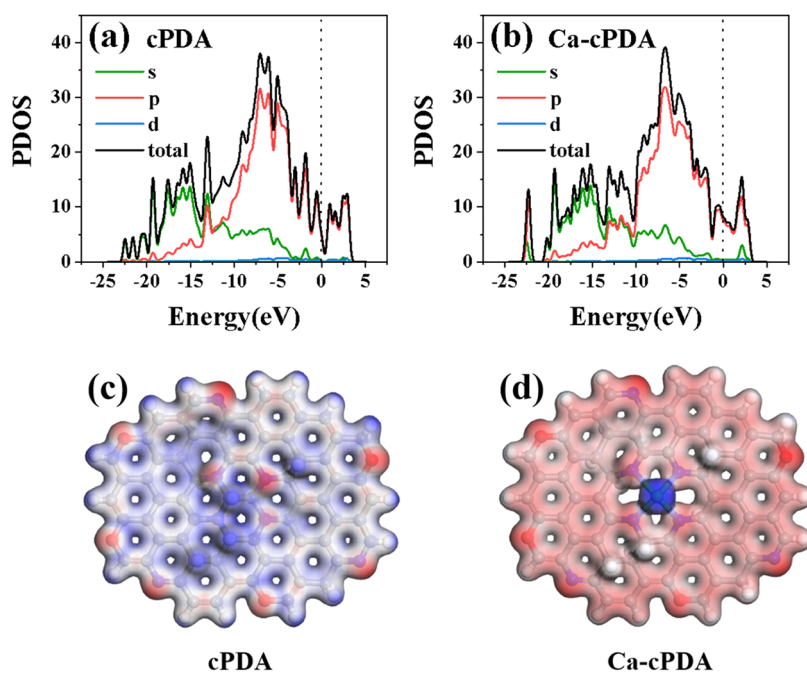


Figure 5. Theoretical modeling of cPDA. Calculated partial density of states (PDOS) (a, b) and electrostatic potential surface (c, d) of graphite-like molecule coordinated without or with Ca^{2+} .

improvement in crystallinity due to the catalytic action of the Fe nanocrystals.

The Raman spectra and XPS results were recorded to further analyze the carbonization process of cPDA, as shown in Figure 4. Figure 4a shows that the graphite-like characteristic peaks near the D-band (1350 cm^{-1}) and G-band (1580 cm^{-1}) are observed for all PDA and metal-PDA. During high-temperature pyrolysis, the amorphous carbon undergoes a structural transformation into short-range ordered graphitic domains, which enhances the G-band intensity associated with in-plane sp^2 hexatomic ring vibrations. Concurrently, the carbonization process generates internanocrystalline defects (e.g., edge dislocations, vacancy clusters) and interfacial boundaries between graphitic microdomains, activating the defect-sensitive D-band. Therefore, after undergoing the carbonization process, PDA exhibits a stronger intensity of the D and G bands.³⁷ The calculated $I_{\text{D}}/I_{\text{G}}$ and G-band shift values are present in Figure 4b, the elevated $I_{\text{D}}/I_{\text{G}}$ value reflects increased internanocrystalline defect density in the carbonization process from amorphous carbon to nanocrystalline graphitic domain, while the enhanced G-band blue shift arises from the increased vibrational frequency of sp^2 -hybridized carbon bonds caused by lattice strain generated during the recrystallization process.^{37–39} Thus, the larger $I_{\text{D}}/I_{\text{G}}$ and G-band blue shift values indicate a more intense graphitization process from amorphous carbon to nanocrystalline graphitic domains during high-temperature annealing in cPDA.

The XPS C 1s spectra are recorded in Figure S4(a–c). The C 1s peak was deconvoluted into four peaks: sp^2 C, sp^3 C, C–O, and C=O.¹⁷ The calculated sp^2 C relative atomic percentages are shown in Figure 4c. The enhanced sp^2 C network is directly related to improving the graphite nanocrystals. As shown in Figure S4(d–f), the deconvoluted N 1s spectra reveal two dominant nitrogen configurations: pyridinic nitrogen (398.6 eV) and graphitic nitrogen (401.1 eV).^{19,40} The proportion of graphitic nitrogen extracted from N 1s spectra gradually increases with an enhancement of the

graphitization degree (Figure 4d). XPS and Raman spectroscopic analyses further reveal the carbonization process of cPDA during the high-temperature annealing and confirm the catalytic role of Fe nanocrystals in graphitic structure formation.

3.3. Ca Coordination Effect on the Electronic Structure of cPDA. To mechanistically elucidate the metal ion's impact on thermoelectric performance, density functional theory (DFT) calculations were conducted to probe electronic structure evolution. Figure S5 shows the graphite nanocrystalline structure section of the cPDA and Ca-cPDA. The density of states (DOS) of the nitrogen-containing graphene-like molecules shown in Figure 5a,b reveals that the valence bands of both systems are predominantly governed by p-orbital electrons. Furthermore, the p-orbitals coordinated with Ca ions exhibit a higher density of states near the Fermi level and a broader DOS distribution. These characteristics, respectively, enhance carrier concentration and improve carrier mobility in the local region. Figure 5c,d shows the electrostatic potential surface of a graphite-like molecule coordinated without or with Ca^{2+} . The red region represents the positive electrostatic potential region, and the blue region represents the negative electrostatic potential region. The extensive negative electrostatic potential arises from π - π -conjugated electron delocalization. Ca^{2+} coordination induces a charge polarization effect, where the Ca–O coordination sites exhibit a higher negative potential in the central region and a positive potential in the peripheral regions. This distortion of the electrostatic potential creates a self-built electric field, driving free electrons in the Ca-coordinated regions to migrate toward uncoordinated areas, thereby enhancing carrier mobility. There is a mechanism of improved electronic transport in graphene nanostructures via Ca^{2+} coordination: the Ca–O coordination introduces extra conducting electrons, while the charge polarization effect lowers the resistance for these electrons to migrate into uncoordinated regions, enhancing local carrier mobility.

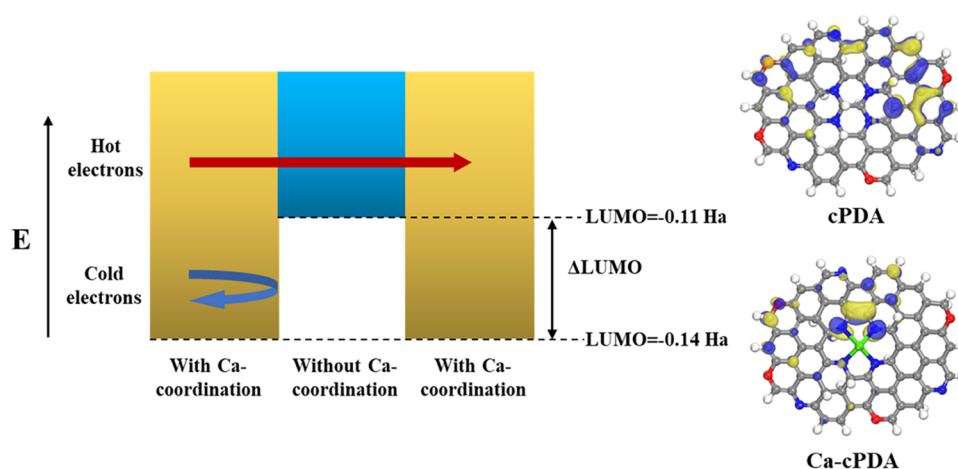


Figure 6. LUMO orbit of graphite-like molecule and energy-filtering effect.

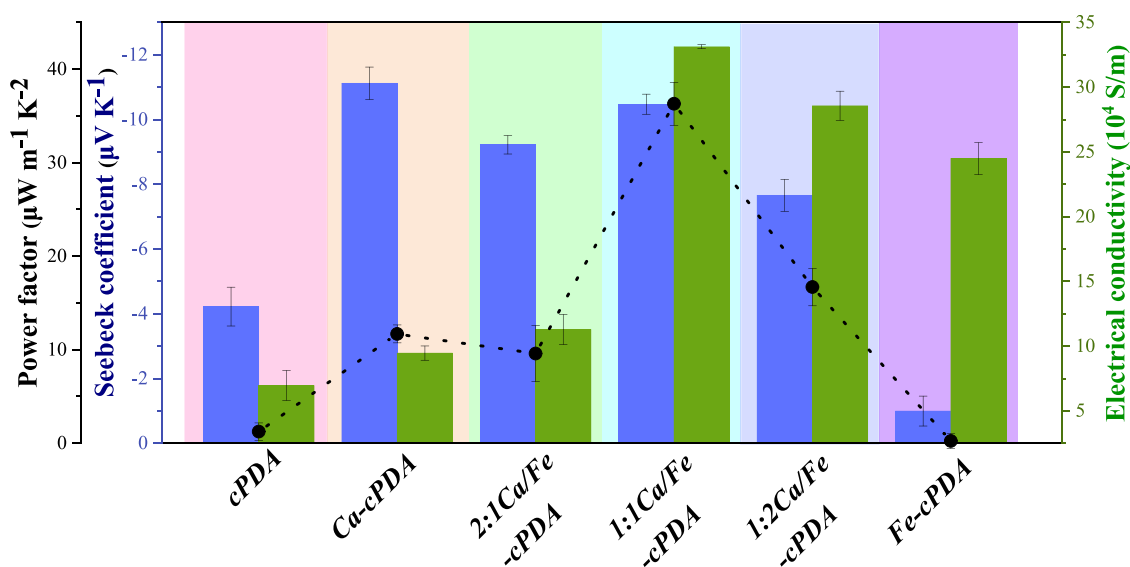


Figure 7. Thermoelectric properties of different cPDA materials. Seebeck coefficient, electrical conductivity, and power factor of cPDA synthesized with different $\text{Ca}^{2+}/\text{Fe}^{3+}$ molar ratios at 300 K.

Figure S6 displays the highest occupied molecular orbital–lowest unoccupied molecular orbital (HOMO–LUMO) orbital diagrams. After Ca coordination, the LUMO energy level decreases, leading to a narrower HOMO–LUMO energy gap. The DOS analysis indicates that Ca coordination provides additional electrons. In n-type semiconductors, where electron transport relies on the LUMO orbitals,⁴¹ this coordination triggers an energy-filtering effect for electrons migrating from Ca-coordinated regions to uncoordinated regions,⁴² as schematically illustrated in Figure 6. Therefore, Ca ion coordination not only increases the concentration of charge carriers but also selectively filters out low-energy electrons through the energy-filtering effect, thereby supplying high-energy electrons. This mechanism enables a significant enhancement of S in cPDA.^{42–45}

3.4. Thermoelectric Property Analysis. Figure 7 presents the thermoelectric performance of the cPDA and $\text{Ca}^{2+}/\text{Fe}^{3+}$ -modified cPDA materials at 300 K. Table S2 lists the thermoelectric properties (S , σ , PF) of all of the tested cPDA samples. Fe catalysis induced remarkable σ enhancement to $2.45 \times 10^5 \text{ S m}^{-1}$ (Fe-cPDA), representing a significant improvement compared to pristine cPDA ($6.9 \times$

10^4 S m^{-1}). XRD analysis (Figure 3b) and TEM characterization (Figure 2d) reveal that this improvement originates from the Fe-mediated development of interconnected graphite microcrystalline networks during pyrolysis. Raman spectroscopy (Figure 4b) further confirms structural evolution, showing an increased I_D/I_G ratio from 0.68 to 1.2 that correlates with the expanded sp^2 domains and optimized carrier transport pathways. Meanwhile, Ca-coordinated cPDA also shows a moderate improvement in σ of approximately $9.4 \times 10^4 \text{ S/m}$. XRD (Figure 3a) and Raman (Figure 4a) analysis show that this enhancement is due to slightly improved graphitization from the Ca-induced carbon layer and higher carrier concentration and carrier mobility from the Ca coordination.⁴⁶ The significant S enhancement from $-4.22 \mu\text{V K}^{-1}$ (cPDA) to $-11.12 \mu\text{V K}^{-1}$ (Ca-cPDA) primarily arises from Ca^{2+} coordination extracting high-energy electrons. The synergistic Ca/Fe system with 1:1 molar ratio (1:1 Ca/Fe-cPDA) achieves concurrent optimization of electrical and thermal transport properties, demonstrating a record σ of $3.31 \times 10^5 \text{ S m}^{-1}$ with a high S of $-10.47 \mu\text{V K}^{-1}$. This concurrent optimization results in a PF of $-36.3 \mu\text{W m}^{-1} \text{ K}^{-2}$, outperforming pristine cPDA ($1.24 \mu\text{W m}^{-1} \text{ K}^{-2}$) by nearly

30 times. Additionally, the thermal conductivities and ZT values of cPDA synthesized with the different ions are represented in Figure S7. The prepared cPDA demonstrated low κ , confirming effective phonon scattering at the graphite/amorphous interfaces.

Unlike conventional n-type polymers that rely on fully conjugated molecular chains, cPDA derives its conductivity from graphitic nanocrystals, which holds promise for enhanced thermal stability. Figure S8 shows the electrical conductivities of Ca/Fe-cPDA versus the tested temperature. The prepared cPDA exhibits a positive temperature dependence (i.e., conductivity increases with rising temperature). The material consists of an amorphous matrix embedded with graphite-like nanocrystals. Notably, traditional graphite or impurity-doped graphite relies on band transport for conduction, typically showing a negative temperature dependence (conductivity decreases as temperature increases). Consequently, the positive temperature dependence observed in cPDA indicates that charge carriers must overcome an interfacial energy gap during transport between the different graphite nanocrystals, suggesting that the conduction relies on quantum tunneling or hopping mechanisms. To probe the thermal degradation threshold, we conducted Raman spectroscopy (Figure S9) under a combined thermal humidity atmosphere (25–600 °C, 60% RH, 30 min exposure). The material maintains structural integrity below 500 °C with characteristic D and G bands. Structural collapse occurred only at 600 °C through carbothermal decomposition of residual amorphous carbon, as verified by near-complete G-band disappearance. This thermal stability matches that of Bi₂Te₃-based counterparts (phase transition at 585 °C) and is comparable to that of other inorganic counterparts (Table S3).

4. CONCLUSIONS

In this work, we developed a strategy synergizing calcium coordination and iron nanoparticle catalysis during the carbonization process of PDA to enhance the thermoelectric performance of cPDA. Fe³⁺ acts as a catalyst during pyrolysis, promoting the formation of graphitic nanocrystals via in situ reduction to metallic Fe nanoparticles, substantially improving conductivity σ by establishing interconnected conductive pathways. At the same time, Ca²⁺ coordination creates energy-filtering barriers that preferentially scatter low-energy carriers, which elevates the S. DFT calculations further validate that Ca²⁺ coordination reduces the HOMO–LUMO gap and modifies the density of states near the Fermi level, facilitating carrier excitation and transport. The bimetallic Ca/Fe-cPDA system intermingles these advantages, resulting in a high σ ($3.31 \times 10^5 \text{ S m}^{-1}$) and enhanced S ($-10.47 \text{ } \mu\text{V K}^{-1}$), with a superior power factor of $36.3 \text{ } \mu\text{W m}^{-1} \text{ K}^{-2}$. The outstanding thermoelectric properties and stability of bimetallic Ca/Fe-cPDAs make them promising for flexible wearable electronics and extreme-environment energy harvesting, especially in self-powered health monitors utilizing the thermal gradient of the human body and in aerospace systems requiring high temperature resistance. Future exploration of coordination-engineered carbon systems should unlock enhanced power factors in excess of $50 \text{ } \mu\text{W m}^{-1} \text{ K}^{-2}$ through targeted density of state optimization.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsaem.5c01945>.

Thermoelectric characterization, thermal conductivity measurement, and computational methods; thermoelectric properties (S , σ , PF) of all the tested cPDA samples; thermal stability and thermal conductivity of typical composites; XPS, TEM, HRTEM images, and Raman spectra; structure diagrams of cPDA and Ca-cPDA in DFT calculation; thermal conductivities and ZT values of cPDA synthesized with the different ions; and electrical conductivities versus the tested temperature (PDF)

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Notes

The authors declare no competing financial interest.

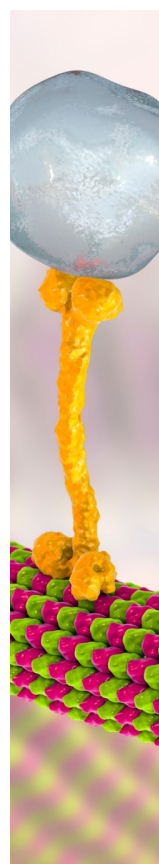
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